

Student's Name:

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Student Number:

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Teacher's Name:

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ABBOTSLEIGH

**2023**

**HIGHER SCHOOL CERTIFICATE**

**Assessment 4**

**Trial Examination**

# Mathematics Extension 1

## General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen.
- **NESA approved** calculators may be used.
- **NESA approved** reference sheet is provided.
- All necessary working should be shown in every question.
- Write your NESA number on each page.
- Start a new page for each of Questions 11 - 14.
- Answer the Multiple Choice questions on the answer sheet provided.
- If you do not attempt a whole question, you must write the question number at the top of a blank page with your NESA number and the words '**NOT ATTEMPTED**' written clearly on it.

**Total marks - 70**

- Attempt Sections I and II.

Section I

Pages 3 - 7

**10 marks**

- Attempt Questions 1–10
- Allow about 15 minutes for this section.

Section II

Pages 8- 16

**60 marks**

- Attempt Questions 11 – 14.
- Allow about 1 hour and 45 minutes for this section.
- All questions are of equal value.

## **Outcomes to be assessed:**

### **Year 11 Mathematics Extension 1 outcomes**

A student:

#### **ME11-1**

uses algebraic and graphical concepts in the modelling and solving of problems involving functions and their inverses

#### **ME11-2**

manipulates algebraic expressions and graphical functions to solve problems

#### **ME11-3**

applies concepts and techniques of inverse trigonometric functions and simplifying expressions involving compound angles in the solution of problems

#### **ME11-4**

applies understanding of the concept of a derivative in the solution of problems, including rates of change, exponential growth and decay and related rates of change

#### **ME11-5**

uses concepts of permutations and combinations to solve problems involving counting or ordering

#### **ME11-7**

communicates making comprehensive use of mathematical language, notation, diagrams and graphs

### **Year 12 Mathematics Extension 1 outcomes**

A student:

#### **ME12-1**

applies techniques involving proof or calculus to model and solve problems

#### **ME12-2**

applies concepts and techniques involving vectors and projectiles to solve problems

#### **ME12-3**

applies advanced concepts and techniques in simplifying expressions involving compound angles and solving trigonometric equations

#### **ME12-4**

uses calculus in the solution of applied problems, including differential equations and volumes of solids of revolution

#### **ME12-7**

evaluates and justifies conclusions, communicating a position clearly in appropriate mathematical forms

## SECTION I

**10 marks**

**Attempt Questions 1 – 10**

**Allow about 15 minutes for this section**

**Use the multiple-choice answer sheet for Questions 1 - 10**

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1. Given  $f(x) = \sqrt{x} - 3$ , what are the domain and range of  $f^{-1}(x)$ ?

(A)  $x \geq -3, y \geq 0$

(B)  $x \geq -3, y \geq -3$

(C)  $x \geq 0, y \geq 0$

(D)  $x \geq 0, y \geq -3$

2. Given  $\tan \theta = \frac{1}{3}$ , what is the exact value of  $\tan\left(\theta + \frac{\pi}{3}\right)$ ?

(A)  $\frac{\sqrt{3} + 3}{3\sqrt{3} - 1}$

(B)  $\frac{\sqrt{3} - 3}{3\sqrt{3} + 1}$

(C)  $\frac{1 + 3\sqrt{3}}{3 - \sqrt{3}}$

(D)  $\frac{1 - 3\sqrt{3}}{3 + \sqrt{3}}$

**Section I continues on next page**

**SECTION I (Continued)**

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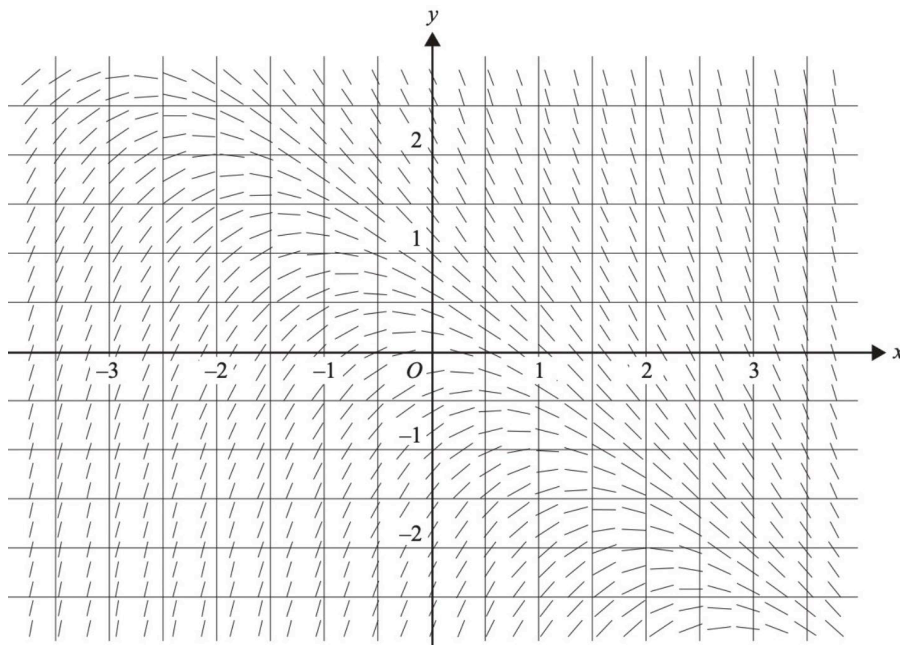
3. What is the domain and range of the function  $y = 6 \sin^{-1}(3x)$ ?
- (A) Domain  $\left[-\frac{1}{3}, \frac{1}{3}\right]$ ; Range  $[-6\pi, 6\pi]$ .
- (B) Domain  $\left[-\frac{1}{3}, \frac{1}{3}\right]$ ; Range  $[-3\pi, 3\pi]$ .
- (C) Domain  $\left[-\frac{1}{3}, \frac{1}{3}\right]$ ; Range  $[0, 6\pi]$ .
- (D) Domain  $\left[-\frac{1}{3}, \frac{1}{3}\right]$ ; Range  $[0, 3\pi]$ .
4. Layla projects an arrow at an angle of  $45^\circ$  to the horizontal with an initial velocity of  $50 \text{ ms}^{-1}$ . What is the horizontal speed of the arrow?
- (A)  $\sqrt{2} \text{ ms}^{-1}$
- (B)  $50\sqrt{2} \text{ ms}^{-1}$
- (C)  $\frac{1}{\sqrt{2}} \text{ ms}^{-1}$
- (D)  $\frac{50}{\sqrt{2}} \text{ ms}^{-1}$

**Section I continues on next page**

**SECTION I (Continued)**

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5. The direction (slope) field for the differential equation  $\frac{dy}{dx} + x + y = 0$  is shown below.



Which point satisfies the solution to the differential equation that includes  $(0, -1)$ ?

- (A)  $(-1.5, -2)$
- (B)  $(2.5, -1)$
- (C)  $(3, -1)$
- (D)  $(3.5, -2.5)$

**Section I continues on next page**

**SECTION I (Continued)**

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6. Which of the following is the solution set of the inequation  $\frac{2}{|x-1|} \leq 3$ ?

(A)  $x < 1$  and  $x \geq \frac{5}{3}$ .

(B)  $1 < x \leq \frac{5}{3}$ .

(C)  $x \leq \frac{1}{3}$  and  $x \geq \frac{5}{3}$ .

(D)  $\frac{1}{3} \leq x \leq \frac{5}{3}$ .

7. What are the values of  $p$  for which  $y = e^{px}$  satisfies the differential equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 12y = 0?$$

(A)  $p = -4, p = 3$

(B)  $p = -3, p = 4$

(C)  $p = \pm 3$

(D)  $p = \pm 4$

8. In a Year 12 Mathematics class, the teacher can give one of 5 grades (A, B, C, D or E) to each student. What is the minimum number of students required so that six students are guaranteed to receive the same grade?

(A) 6

(B) 25

(C) 26

(D) 30

**Section I continues on next page**

**SECTION I (Continued)**

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9. In how many different ways can three girls and three boys be seated on a bench for a photograph if the girls must sit apart?

- (A) 72
- (B) 120
- (C) 144
- (D) 576

10. Which of the following is equal to  $\int_{-1}^1 \cos^{-1}x \, dx$ ?

- (A)  $\frac{\pi}{3}$
- (B)  $\frac{\pi}{2}$
- (C)  $\pi$
- (D)  $2\pi$

**End of Section I**

## SECTION II

**60 marks**

**Attempt Questions 11 – 14**

**Allow about 1 hour and 45 minutes for this section**

Answer each question in a **SEPARATE** writing booklet. Extra writing booklets are available.

In Questions 11-14 your responses should include relevant mathematical reasoning and/or calculations.

**Marks**

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### Question 11 (15 marks)

- (a) The polynomial  $P(x) = 8x^4 - 38x^3 + 9x^2 + ax + b$  has a double root at  $x = 3$ . **3**

Find the value of  $a$  and  $b$ , where  $a$  and  $b$  are real numbers.

- (b) Find the exact value of  $\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$ . **2**

- (c) Consider the expansion of  $(2x - p)^9$ . The coefficient of the term in  $x^6$  is  $-672\,000$ . **2**

Find the value of  $p$ .

**Question 11 continues on next page**

**QUESTION 11 (Continued)****Marks**

- 
- (d) (i) Write the expression  $2\sqrt{3} \sin x + 2 \cos x$  in the form  $R \sin(x + \alpha)$ , **2**  
where  $R > 0$  and  $0 \leq \alpha \leq \frac{\pi}{2}$ .

- (ii) Hence find the value of  $k$ , where  $0 \leq k \leq 2\pi$ , for which **3**

$$\int_0^k (2\sqrt{3} \cos x - 2 \sin x) \, dx = 2.$$

- (e) Consider the function  $f(x) = x^2 - c^2$ , where  $c$  is a positive real number. **3**

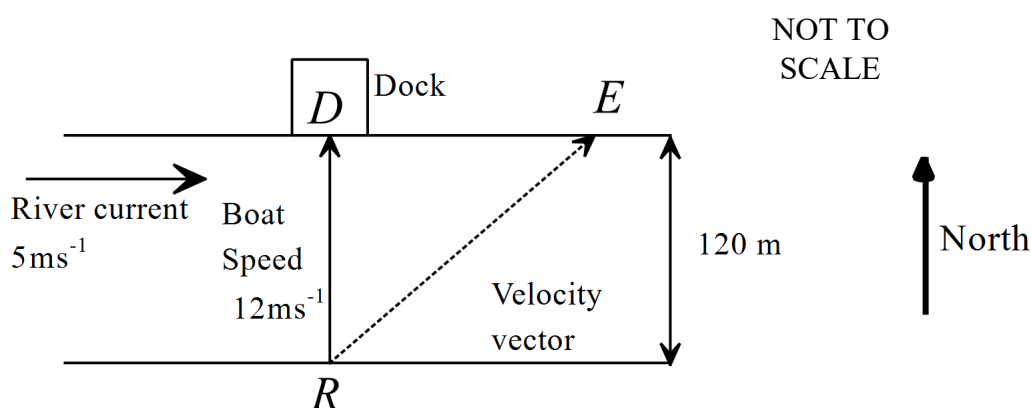
Sketch the graph of  $y = \frac{1}{|f(x)|}$ , showing all important features including the turning point(s), intercept(s) and asymptote(s).

**End of Question 11**

(a) Find  $\int \frac{\cos^2(\ln x)}{x} dx$  using the substitution  $u = \ln x$ . 3

- (b) Rylie has a boat which moves at a top speed of  $12 \text{ ms}^{-1}$  in still water.

From point  $R$ , he wants to go due north to point  $D$  on the opposite side of the river, as shown in the diagram below.



Today the current in the river is flowing at  $5 \text{ ms}^{-1}$ . From  $R$ , he steers the boat due north toward  $D$  at top speed. Due to the current he drifts down the river and arrives at point  $E$ .

- (i) Taking  $R$  as the origin, write down Rylie's velocity vector in the form  $x\mathbf{i} + y\mathbf{j}$  and find the magnitude of this vector. 2
- (ii) What is the bearing of Rylie's velocity vector and how far does he travel from  $R$  to  $E$ ? 2
- (iii) On what bearing should Rylie have pointed the boat, so that he arrived at  $D$ , with the boat travelling at top speed? 2

Question 12 continues on next page

(c) (i) Show that  $\frac{1}{(n+1)!} - \frac{n+1}{(n+2)!} = \frac{1}{(n+2)!}$  1

(ii) Use mathematical induction to show that, for all integers  $n \geq 1$ , 3

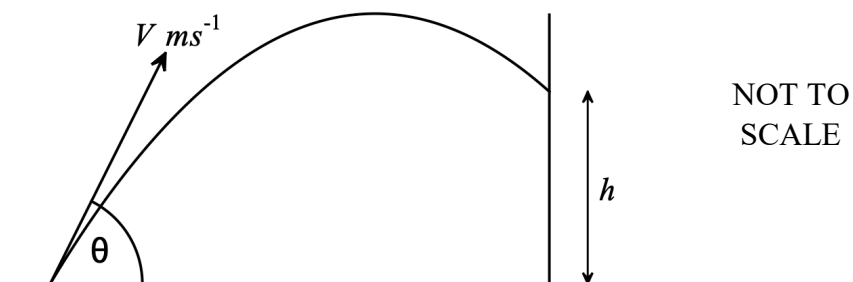
$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}.$$

(d) The roots of  $x^3 + 4x^2 - 5x + 3 = 0$  are  $\alpha$ ,  $\beta$ , and  $\gamma$ . 2

Find the value of  $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$ .

**End of Question 12**

- (a) A rocket is fired with a velocity  $V \text{ ms}^{-1}$  at an angle  $\theta$  to the horizontal from sea level and hits a cliff face as shown. Neglect air resistance and take  $g = 10 \text{ ms}^{-2}$ .



- (i) If  $V = 180 \text{ ms}^{-1}$  and  $\theta = 60^\circ$ , show that at any time  $t$  seconds, the position vector of the rocket relative to the point at which it was fired is 2

$$\underline{s}(t) = 90t\underline{i} + (90\sqrt{3}t - 5t^2)\underline{j}$$

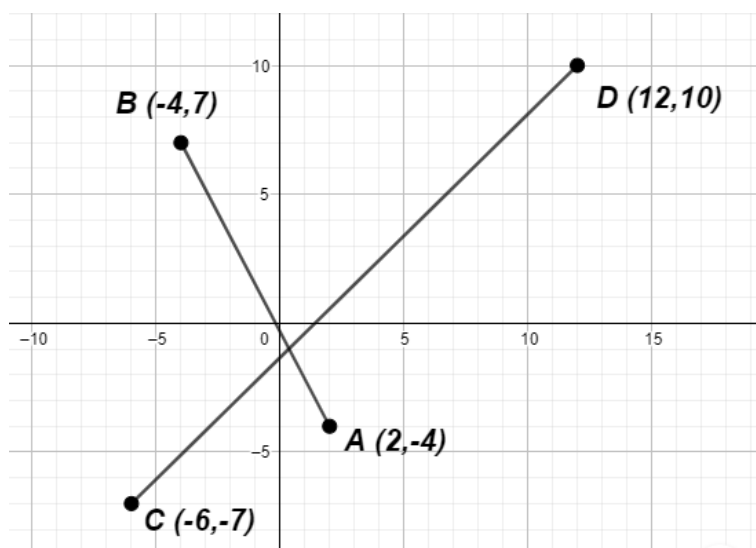
- (ii) Find the maximum height reached by the rocket. 2
- (iii) The base of the cliff is 270 m from the point at which the rocket was fired. Find the value of  $h$ . 2

Question 13 continues on next page

- (b) The diagram below shows the lines  $AB$  and  $CD$ . The line  $AB$  passes through the Points  $(2, -4)$  and  $(-4, 7)$  and the line  $CD$  passes through  $(-6, -7)$  and  $(12, 10)$ . 3

Use vector methods to find the acute angle between the two lines.

Write your answer to the nearest degree.



- (c) The proportion,  $P$ , of people who know a rumour is modelled by the differential equation  $\frac{dP}{dt} = 2P(1 - P)$  where  $t$  is the time in weeks. Initially only 10% of people know the rumour.

- (i) Show that  $\frac{1}{P} + \frac{1}{1-P} = \frac{1}{P(1-P)}$ . 1
- (ii) Solve the differential equation for  $P$ . 3
- (iii) Hence, find the exact time taken for half of the people to know the rumour. 2

End of Question 13

- (a) Find a cartesian equation for the following pair of parametric equations:

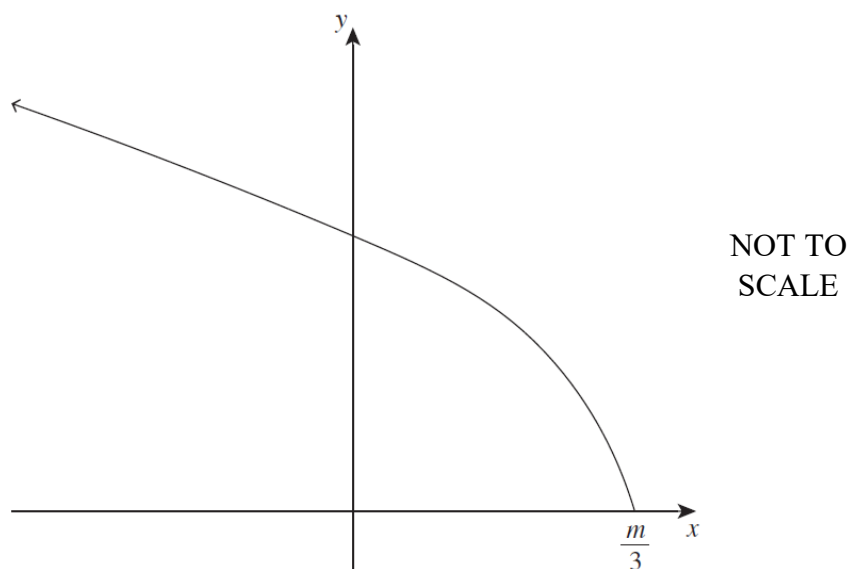
2

$$x = 5 \sin \theta$$

$$y = 5 \cos \theta + 1$$

- (b) Let  $f(x) = \sqrt{m - 3x}$  for  $x \leq \frac{m}{3}$ . The graph of  $f(x)$  is shown below.

4



The area enclosed by the graph  $f(x)$ , the  $x$ -axis and the  $y$ -axis is rotated about the  $y$ -axis. Find the value of  $m$  such that the volume of the solid formed is  $\frac{5000\pi}{27}$  units<sup>3</sup>.

Question 14 continues on next page

- (c) The velocity of a particle moving in a straight line is given by  $\dot{x} = e^x + e^{-x}$ , where  $x$  in metres is the displacement from the origin. Initially the particle is at the origin.

- (i) Show that the time taken by the particle as a function of its displacement is given by 2

$$t = \int \frac{e^x}{e^{2x} + 1} dx.$$

- (ii) Hence, by using the substitution  $u = e^x$ , prove that. 2

$$x = \ln \left( \tan \left( t + \frac{\pi}{4} \right) \right).$$

- (d) (i) Show that  $\tan^{-1}(n+1) - \tan^{-1}(n-1) = \tan^{-1}\left(\frac{2}{n^2}\right)$  for  $n \geq 1$ . 2

- (ii) Hence, or otherwise, show that: 3

$$\sum_{r=1}^n \tan^{-1}\left(\frac{2}{r^2}\right) = \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) - \frac{\pi}{4}.$$

**End of Paper**

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# Ext1 Trial 2023 Solutions

## Multiple choice

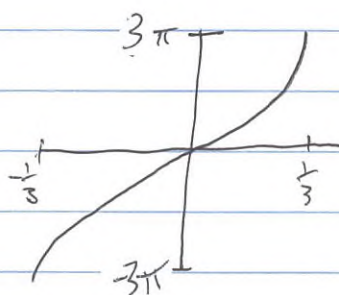
1.  $f(x) = \sqrt{x} - 3$  Dom:  $x \geq 0$ , Range  $y \geq -3$

$\therefore f^{-1}(x)$  Domain:  $x \geq -3$ , Range  $y \geq 0 \therefore (A)$

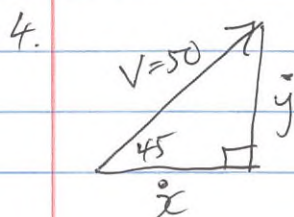
2.  $\tan\left(\theta + \frac{\pi}{3}\right) = \frac{\tan\theta + \tan\frac{\pi}{3}}{1 - \tan\theta \tan\frac{\pi}{3}}$

$= \frac{\frac{1}{3} + \sqrt{3}}{1 - \frac{\sqrt{3}}{3}} = \frac{(1 + 3\sqrt{3})/3}{(3 - \sqrt{3})/3} \therefore (C)$

3.  $y = 6 \sin^{-1}(3x)$



$\therefore (B)$

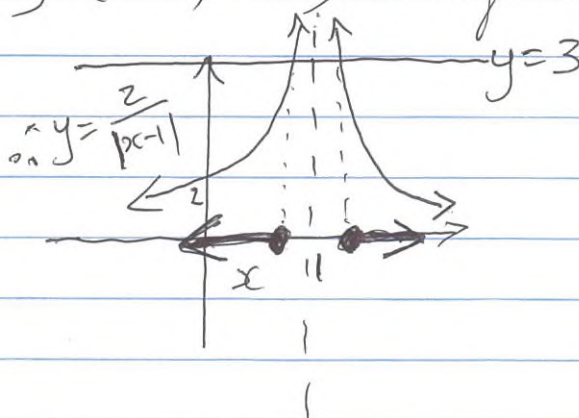
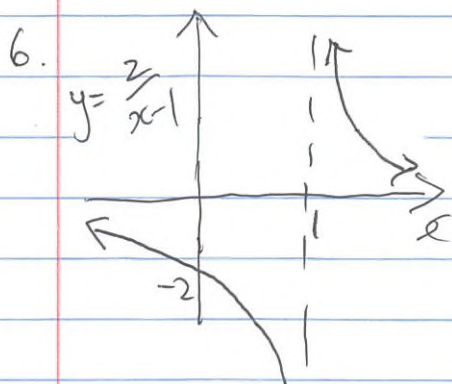


$\frac{1}{\sqrt{2}} = \frac{\dot{x}}{50}$

$\therefore \dot{x} = \frac{50}{\sqrt{2}}$

$\therefore (D)$

5. By inspection, only  $(3.5, -2.5)$  satisfies the DE  $\therefore (D)$



$\therefore (C)$

7.  $y' = p e^{px}$   
 $y'' = p^2 e^{px}$

i.e.  $p^2 e^{px} - p e^{px} - 12 e^{px} = 0$   
 $e^{px} (p^2 - p - 12) = 0$   
 $e^{px} (p-4)(p+3) = 0$

$\therefore (B)$

8.  $5A's + 5B's + 5C's + 5D's + 5E's + 1$   
 $= 26$

$\therefore (C)$

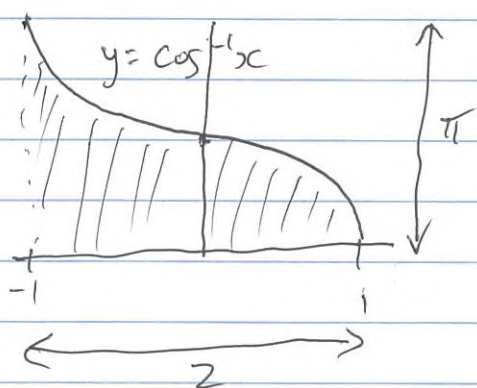
9.  $B_1, B_2, B_3 \rightarrow 3!$  ways to arrange the group of boys.

i.e.  $B_1, B_2, B_3 \rightarrow 4!$  ways to arrange the girls.

$\therefore 3! \times 4!$

$\therefore (C)$

10.



$\int_{-1}^1 \cos^{-1}x \, dx = \frac{1}{2} \cdot 2\pi \therefore (C)$

## Question 11

$$(a) \quad P(x) = 8x^4 - 38x^3 + 9x^2 + ax + b$$

$$P'(x) = 32x^3 - 114x^2 + 18x + a$$

$$\text{now } P(3) = 648 - 1026 + 81 + 3a + b = 0$$

$$\text{i.e. } 3a + b = 297$$

$$\text{and } P'(3) = 864 - 1026 + 54 + a = 0$$

$$\underline{a = 108}$$

$$\therefore 3(108) + b = 297$$

$$\underline{b = -27}$$

$$(b) \quad \int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{2^2 - x^2}} dx = \left[ \sin^{-1}\left(\frac{x}{2}\right) \right]_{\sqrt{2}}^{\sqrt{3}}$$

$$= \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$= \frac{\pi}{3} - \frac{\pi}{4}$$

$$= \frac{\pi}{12}$$

$$(c) \quad (2x - p)^9 = \sum_{k=0}^9 \binom{9}{k} (2x)^k (-p)^{9-k}$$

$$\text{term in } x^6 = \binom{9}{6} (2x)^6 (-p)^3$$

$$\text{i.e. } -672000 = -84 \cdot 64 \cdot p^3$$

$$p^3 = 125$$

$$p = 5$$

$$d(i) \quad 2\sqrt{3} \sin x + 2\cos x = R \sin x \cos \alpha + R \sin \alpha \cos x$$

$$R \cos \alpha = 2\sqrt{3}$$

$$R \sin \alpha = 2$$

$$R^2 (\cos^2 \alpha + \sin^2 \alpha) = (4 \times 3) + 4$$

$$\therefore R = 4 \quad (R > 0)$$

$$\frac{R \sin \alpha}{R \cos \alpha} = \frac{2}{2\sqrt{3}} = \tan \alpha$$

$$\text{i.e. } \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6} \quad (0 < \alpha < \frac{\pi}{2})$$

$$\text{i.e. } 2\sqrt{3} \sin x + 2\cos x = 4 \sin \left(x + \frac{\pi}{6}\right)$$

$$(ii) \quad \int_0^k (2\sqrt{3} \cos x - 2 \sin x) dx$$

$$= [2\sqrt{3} \sin x + 2 \cos x]_0^k = 2$$

$$\text{i.e. } 2\sqrt{3} \sin k + 2 \cos k - (2\sqrt{3} \sin 0 + 2 \cos 0) = 2$$

$$2\sqrt{3} \sin k + 2 \cos k - 2 = 2$$

$$\text{i.e. } 2\sqrt{3} \sin k + 2 \cos k = 4$$

$\therefore$  using (i)

$$4 \sin \left(k + \frac{\pi}{6}\right) = 4$$

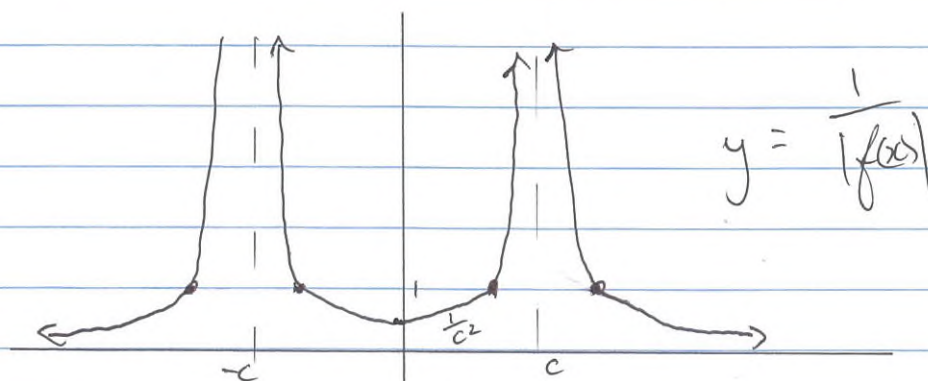
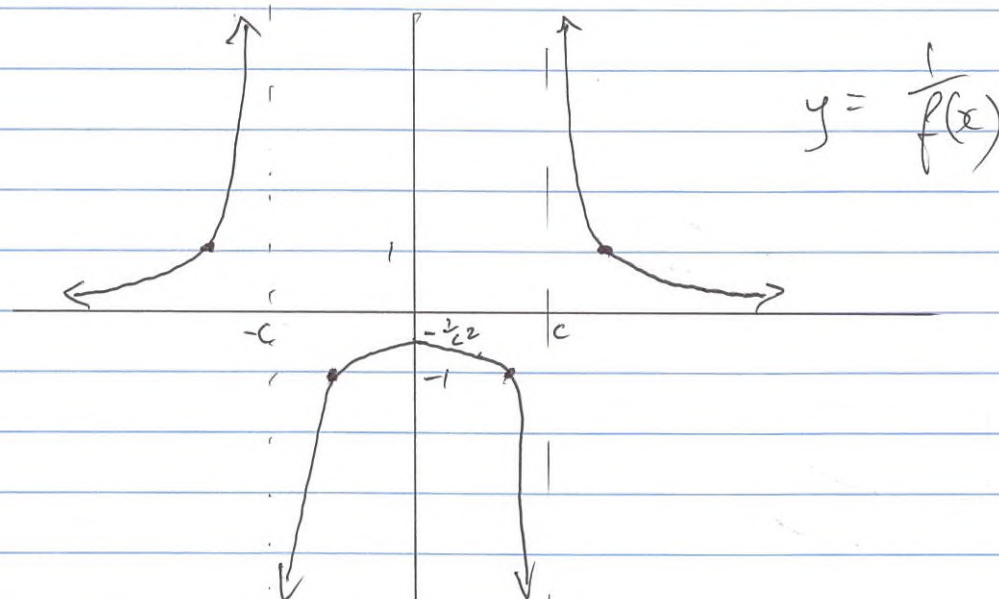
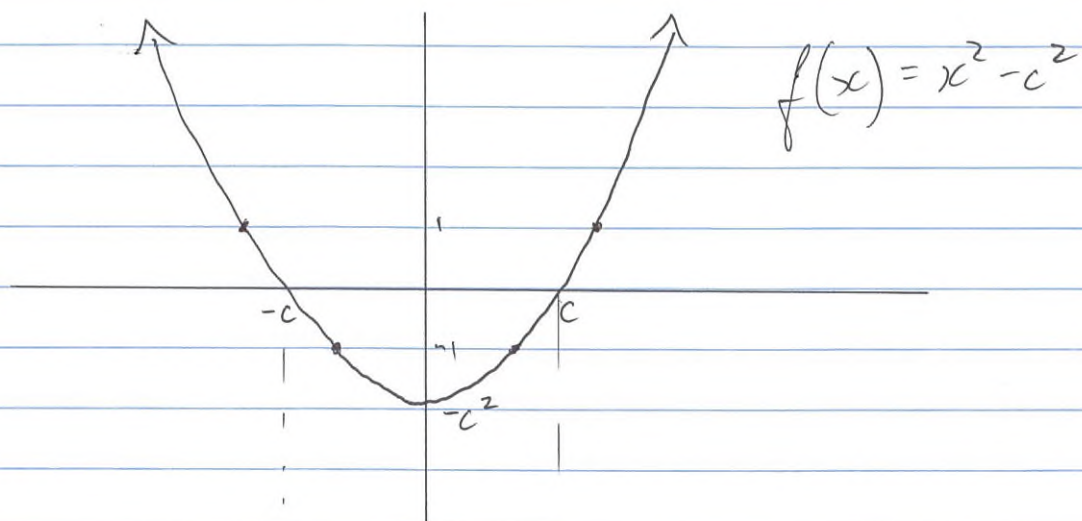
$$\sin \left(k + \frac{\pi}{6}\right) = 1$$

$$k + \frac{\pi}{6} = \frac{\pi}{2}, \frac{5\pi}{2}, \dots$$

$$\therefore k = \frac{\pi}{3}$$

1 solution only for  $0 \leq k \leq 2\pi$

(e)



asymptotes @  $x = \pm c$   
turning point and y-intercept @  $(0, \frac{1}{c^2})$

## Question 12.

(a)  $I = \int \frac{\cos^2(\ln x)}{x} dx$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$\therefore I = \int \cos^2(\ln x) \frac{dx}{x} = \int \cos^2 u du$$

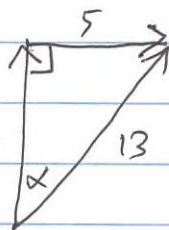
$$= \frac{1}{2} \int 1 + \cos 2u du$$

$$= \frac{1}{2} \left[ u + \frac{1}{2} \sin 2u \right]$$

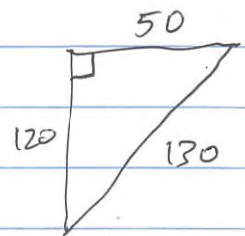
$$\therefore I = \frac{1}{2} \left[ \ln x + \frac{1}{2} \sin(2 \ln x) \right] + C$$

(b) (i)  $\vec{v} = 5\vec{i} + 12\vec{j}$

Using Pythagoras',  $|\vec{v}| = 13 \text{ m/s}$



(ii) From the diagram,  $\tan \alpha = \frac{5}{12}$   
 $\therefore \alpha = 22.61^\circ$

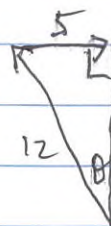


$\therefore$  Rylie's bearing is  $022.6^\circ$  and he travels 130m.

(iii) using the triangle:

$$\sin \theta = \frac{5}{12}$$

$$\therefore \theta = 24.62^\circ$$



i.e Rylie should use a bearing of  $360 - 24.62$   
 $\therefore 335^\circ$

$$\begin{aligned}
 12(c)(i) \text{ LHS} &= \frac{1}{(n+1)!} - \frac{n+1}{(n+2)!} \\
 &= \frac{(n+2) - (n+1)}{(n+2)!} \\
 &= \frac{1}{(n+2)!} = \text{RHS}.
 \end{aligned}$$

(ii) Prove  $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$  for  $n \geq 1$

Test  $n=1$

$$\begin{aligned}
 \text{LHS} &= \frac{1}{(1+1)!} = \frac{1}{2!} = \frac{1}{2} \\
 \text{RHS} &= 1 - \frac{1}{(1+1)!} \\
 &= 1 - \frac{1}{2!} \\
 &= \frac{1}{2} = \text{LHS}.
 \end{aligned}$$

$\therefore$  true for  $n=1$

Assume true for  $n=k$ , i.e. assume:

$$\frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$$

Now show true for  $n=k+1$ , i.e. show:

$$\frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} = 1 - \frac{1}{(k+2)!}$$

$$\begin{aligned}
 \text{LHS} &= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \quad \text{using the assumption} \\
 &= 1 - \left( \frac{1}{(k+1)!} - \frac{k+1}{(k+2)!} \right) \\
 &= 1 - \frac{1}{(k+2)!} \quad \text{using part (i)} \\
 &= \text{RHS}.
 \end{aligned}$$

$$(d) \quad x^3 + 4x^2 - 5x + 3 = 0$$

Now

$$\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$$

$$= \frac{\gamma}{\alpha\beta\gamma} + \frac{\beta}{\alpha\beta\gamma} + \frac{\alpha}{\alpha\beta\gamma}$$

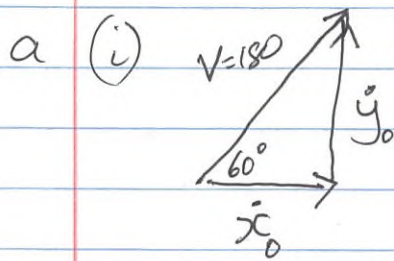
$$= \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma}$$

$$= \frac{-b/a}{-d/a}$$

$$= -4 / -3$$

$$= 4/3$$

### Question 13



initial velocity components:

$$\begin{aligned}\dot{x}_0 &= 180 \cos 60 \\ &= 90 \text{ m/s}\end{aligned}$$

$$\begin{aligned}\dot{y}_0 &= 180 \sin 60 \\ &= 90\sqrt{3} \text{ m/s}.\end{aligned}$$

horizontal motion:

$$\begin{aligned}\ddot{x} &= 0 \\ \dot{x} &= \int 0 \, dt \\ &= C\end{aligned}$$

at  $t=0$ ,  $\dot{x} = 90$

$$\begin{aligned}\therefore x &= \int_0^t 90 \, dt \\ &= [90t]_0^t \\ &= 90t - 0\end{aligned}$$

$x = 90t$

vertical motion:

$$\begin{aligned}\ddot{y} &= -10 \\ \dot{y} &= \int -10 \, dt \\ &= -10t + C\end{aligned}$$

at  $t=0$ ,  $\dot{y} = 90\sqrt{3}$

$\therefore \dot{y} = -10t + 90\sqrt{3}$

$$\begin{aligned}\therefore y &= \int -10t + 90\sqrt{3} \, dt \\ &= -5t^2 + 90\sqrt{3}t + C\end{aligned}$$

at  $t=0$ ,  $y=0$ ,  $\therefore C=0$

i.e.  $y = -5t^2 + 90\sqrt{3}t$

$\therefore s(t) = 90t \underline{\hat{i}} + (90\sqrt{3}t - 5t^2) \underline{\hat{j}}$

(ii) Maximum height occurs when  $\dot{y} = 0$

i.e. when  $90\sqrt{3} - 10t = 0$

$$9\sqrt{3} - t = 0$$

$$t = 9\sqrt{3} \text{ sec}$$

Sub  $t = 9\sqrt{3}$  into  $y$ :

$$\begin{aligned}y &= -5(9\sqrt{3})^2 + 90\sqrt{3}(9\sqrt{3}) \\ &= 1215 \text{ m}.\end{aligned}$$

(iii)  $x = 270 = 90t$

$\therefore$  at  $t = 3 \text{ sec}$  (sub this into  $y$ )

$$\begin{aligned}y &= -5(3)^2 + 90\sqrt{3}(3) \\ &= (-45 + 270\sqrt{3}) \text{ m}\end{aligned}$$

$$13(b) \quad \vec{BA} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} - \begin{pmatrix} -4 \\ 7 \end{pmatrix} = \begin{pmatrix} 6 \\ -11 \end{pmatrix} \quad \therefore |\vec{BA}| = \sqrt{6^2 + 11^2} = \sqrt{157}$$

$$\vec{DC} = \begin{pmatrix} -6 \\ -7 \end{pmatrix} - \begin{pmatrix} 12 \\ 10 \end{pmatrix} = \begin{pmatrix} -18 \\ -17 \end{pmatrix} \quad \therefore |\vec{DC}| = \sqrt{18^2 + 17^2} = \sqrt{613}$$

$$\begin{aligned} \vec{BA} \cdot \vec{DC} &= \begin{pmatrix} 6 \\ -11 \end{pmatrix} \cdot \begin{pmatrix} -18 \\ -17 \end{pmatrix} \\ &= 6 \times (-18) + (-11) \cdot (-17) \\ &= 79 \end{aligned}$$

and

$$\vec{BA} \cdot \vec{DC} = |\vec{BA}| \times |\vec{DC}| \cos \theta$$

i.e

$$79 = \sqrt{157} \cdot \sqrt{613} \cos \theta$$

$$\therefore \cos \theta = \frac{79}{\sqrt{157} \cdot \sqrt{613}}$$

$$\theta = 75.247 \dots$$

i.e  $\theta = 75^\circ$  (to nearest degree).

$$13(c) \quad (i) \quad LHS = \frac{1}{P} + \frac{1}{1-P}$$

$$= \frac{1-P + P}{P(1-P)} = RHS.$$

$$(ii) \quad \frac{dP}{dt} = 2P(1-P)$$

Separate variables:

$$\int_{0.1}^P \frac{dP}{P(1-P)} = 2 \int_0^t dt$$

using part (i)

$$\int_{0.1}^P \frac{1}{P} + \frac{1}{1-P} dP = 2 \left[ t \right]_0^t$$

$$\left[ \ln P - \ln(1-P) \right]_{0.1}^P = 2t - 0$$

$$\ln P - \ln(1-P) - (\ln(0.1) - \ln(0.9)) = 2t$$

$$\ln \frac{P}{1-P} + \ln \frac{9}{1} = 2t$$

$$\text{i.e.} \quad 2t = \ln \frac{9P}{1-P}$$

$$e^{2t} = \frac{9P}{1-P}$$

$$(1-P)e^{2t} = 9P$$

$$e^{2t} = 9P + Pe^{2t}$$

$$= P(9 + e^{2t})$$

$$\therefore P = \frac{e^{2t}}{9 + e^{2t}} \cdot \frac{e^{-2t}}{e^{-2t}} = \frac{1}{9e^{-2t} + 1}$$

$$(iii) \quad \text{When } P = \frac{1}{2}$$

$$\frac{1}{2} = \frac{1}{9e^{-2t} + 1}$$

$$\therefore 2 = 9e^{-2t} + 1$$

$$e^{-2t} = \frac{1}{9}$$

$$e^{2t} = 9$$

$$\therefore t = \frac{1}{2} \ln 9$$

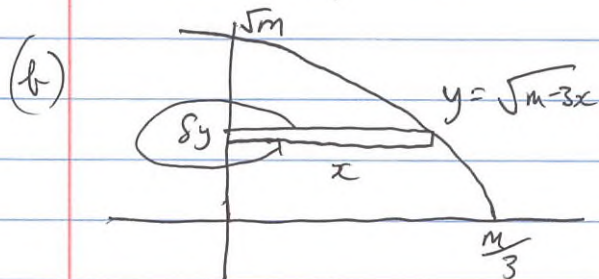
# Question 14

$$(a) \quad \sin \theta = \frac{x}{5}, \quad \cos \theta = \frac{y-1}{5}$$

$$\therefore \sin^2 \theta + \cos^2 \theta = 1$$

$$\text{i.e.} \quad \frac{x^2}{25} + \frac{(y-1)^2}{25} = 1$$

$$\text{alternatively } x^2 + (y-1)^2 = 25$$



$$\begin{aligned} y &= \sqrt{m-3x} \\ y^2 &= m-3x \\ x &= \frac{m-y^2}{3} \end{aligned}$$

$$\therefore x^2 = \frac{m^2 - 2my^2 + y^4}{9}$$

typical slice rotated

will have  $V = \pi x^2 \delta y$

$$\therefore V = \pi \int_0^{\sqrt{m}} x^2 dy$$

$$= \frac{\pi}{9} \int_0^{\sqrt{m}} m^2 - 2my^2 + y^4 dy$$

$$= \frac{\pi}{9} \left[ m^2 y - \frac{2my^3}{3} + \frac{y^5}{5} \right]_0^{\sqrt{m}}$$

$$= \frac{\pi}{9} \left[ m^{2.5} - \frac{2}{3} m^{2.5} + \frac{1}{5} m^{2.5} - 0 \right]$$

i.e.

$$\frac{5000\pi}{27} = \frac{\pi}{9} \left( \frac{8}{15} m^{5/2} \right)$$

$$m^{5/2} = \frac{5000}{27} \cdot \frac{9 \cdot 15}{8}$$

$$m^{5/2} = 3125$$

$$\therefore m = 25$$

$$(c) (i) \dot{x} = \frac{dx}{dt} = e^x + e^{-x} \\ = e^x + \frac{1}{e^x} = \frac{e^{2x} + 1}{e^x}$$

$$\therefore \frac{dt}{dx} = \frac{e^x}{e^{2x} + 1}$$

$$\therefore \int_0^t dt = \int_0^x \frac{e^x}{e^{2x} + 1} dx$$

$$(ii) \text{ Let } u = e^x$$

$$du = e^x dx$$

$$\text{When } x=0, u=1 \\ x=x, u=e^x$$

$$\therefore t = \int_1^{e^x} \frac{du}{u^2 + 1}$$

$$= [\tan^{-1} u]_1^{e^x}$$

$$= \tan^{-1} e^x - \tan^{-1} 1$$

$$\therefore t = \tan^{-1} e^x - \frac{\pi}{4}$$

$$(t + \frac{\pi}{4}) = \tan^{-1} e^x$$

$$e^x = \tan(t + \frac{\pi}{4})$$

$$\therefore x = \ln \left[ \tan \left( t + \frac{\pi}{4} \right) \right]$$

(i) show  $\tan^{-1}(n+1) - \tan^{-1}(n-1) = \tan^{-1}\left(\frac{2}{n^2}\right)$   $n \geq 1$

(d) now,

$$\begin{aligned} \tan [\tan^{-1}(n+1) - \tan^{-1}(n-1)] &= \frac{\tan[\tan^{-1}(n+1)] - \tan[\tan^{-1}(n-1)]}{1 + \tan[\tan^{-1}(n+1)] \tan[\tan^{-1}(n-1)]} \\ &= \frac{n+1 - (n-1)}{1 + (n+1)(n-1)} \\ &= \frac{2}{n^2} \end{aligned}$$

i.e.  $\tan^{-1}(n+1) - \tan^{-1}(n-1) = \tan^{-1}\left(\frac{2}{n^2}\right)$ .

(ii) Hence show  $\sum_{r=1}^n \tan^{-1}\left(\frac{2}{r^2}\right) = \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) - \frac{\pi}{4}$

LHS =  $\sum_{r=1}^n [\tan^{-1}(k+1) - \tan^{-1}(k-1)]$  using part (i)

$$= \tan^{-1}2 - \tan^{-1}0 + \tan^{-1}3 - \tan^{-1}1 + \tan^{-1}4 - \tan^{-1}2 + \tan^{-1}5 - \tan^{-1}3 + \dots$$

$$+ \dots + \tan^{-1}(n-1) - \tan^{-1}(n-3) + \tan^{-1}(n) - \tan^{-1}(n-2) + \tan^{-1}(n+1) - \tan^{-1}(n-1)$$

Rearranging :

$$= [\tan^{-1}2 + \tan^{-1}3 + \tan^{-1}4 + \dots + \tan^{-1}(n-1)] + \tan^{-1}(n) + \tan^{-1}(n+1)$$

$$- [\tan^{-1}(0) + \tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 + \dots + \tan^{-1}(n-2) + \tan^{-1}(n-1)]$$

$\therefore$  Remaining terms after cancelling =  $\tan^{-1}(n) + \tan^{-1}(n+1) - \tan^{-1}1$

Now, similar to part (i):

$$\tan [\tan^{-1}(n) + \tan^{-1}(n+1)] = \frac{n + n+1}{1 - n(n+1)} = \frac{2n+1}{1-n-n^2}$$

i.e.  $\tan^{-1}(n) + \tan^{-1}(n+1) = \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right)$

$$\begin{aligned} \therefore \sum_{r=1}^n \tan^{-1}\left(\frac{2}{r^2}\right) &= \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) - \tan^{-1}1 \\ &= \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) - \frac{\pi}{4} \end{aligned}$$

by induction  $\sum_{r=1}^n \tan^{-1}\left(\frac{2}{r^2}\right) = \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) - \frac{\pi}{4}$

Step 1: Let  $n=1$

$$\text{LHS} = \sum_{r=1}^1 \tan^{-1}\left(\frac{2}{r^2}\right)$$

$$= \tan^{-1}(2)$$

$$\text{RHS} = \tan^{-1}\left(\frac{2(1)+1}{1-1-(1)^2}\right) - \frac{\pi}{4}$$

$$= \tan^{-1}(-3) - \frac{\pi}{4}$$

$$\text{now } \tan\left(\tan^{-1}(-3) - \frac{\pi}{4}\right) = \frac{\tan(\tan^{-1}(-3)) - \tan \frac{\pi}{4}}{1 + \tan(\tan^{-1}(-3)) \tan \frac{\pi}{4}}$$

$$= \frac{-3 - 1}{1 + -3 \times 1}$$

$$= 2$$

$$\therefore \tan^{-1}(-3) - \frac{\pi}{4} = \tan^{-1}(2)$$

$$= \text{RHS}$$

Step 2: Assume true for  $n=k$ .

$$\text{i.e. } \sum_{r=1}^k \tan^{-1}\left(\frac{2}{r^2}\right) = \tan^{-1}\left(\frac{2k+1}{1-k-k^2}\right) - \frac{\pi}{4}$$

Step 3: Prove for  $n=k+1$

$$\text{i.e. } \sum_{r=1}^{k+1} \tan^{-1}\left(\frac{2}{r^2}\right) = \tan^{-1}\left(\frac{2(k+1)+1}{1-(k+1)-(k+1)^2}\right)$$

$$= \tan^{-1}\left(\frac{2k+3}{-k^2-3k-1}\right)$$

$$\text{LHS} = \sum_{r=1}^k \tan^{-1}\left(\frac{2}{r^2}\right) + \tan^{-1}\left(\frac{2}{(k+1)^2}\right)$$

$$= \tan^{-1}\left(\frac{2k+1}{1-k-k^2}\right) + \tan^{-1}\left(\frac{2}{(k+1)^2}\right) \text{ from assumption.}$$

$$\text{So } \tan(\text{LHS}) = \frac{\tan(\tan^{-1}(\frac{2k+1}{1-k-k^2})) + \tan(\tan^{-1}(\frac{2}{(k+1)^2}))}{1 - \tan(\tan^{-1}(\frac{2k+1}{1-k-k^2})) \tan(\tan^{-1}(\frac{2}{(k+1)^2}))}$$

$$= \frac{\frac{2k+1}{1-k-k^2} + \frac{2}{(k+1)^2}}{1 - \frac{2k+1}{1-k-k^2} \cdot \frac{2}{(k+1)^2}}$$

$$= \frac{(2k+1)(k+1)^2 + 2(1-k-k^2)}{(1-k-k^2)(k+1)^2 - 2(2k+1)}$$

$$= \frac{2k^3 + 4k^2 + 2k + k^2 + 2k + 1 + 2 - 2k - 2k^2}{k^2 + 2k + 1 - k^2 - 2k^2 - k - k^4 - 2k^3 - k^2 - 4k - 2}$$

$$= \frac{2k^3 + 3k^2 + 2k + 3}{-k^4 - 3k^3 - 2k^2 - 3k - 1}$$

$$= \frac{(2k+3)(k^2+1)}{-(k^2+1)(k^2+3k+1)}$$

$$\text{as } 2k^3 + 3k^2 + 2k + 3 = (2k+3)k^2 + (2k+3) \\ = (2k+3)(k^2+1)$$

$$= \frac{2k+3}{-k^2-3k-1}$$

$$\begin{array}{r} \phantom{k^2+1} \overline{k^2+3k+1} \\ k^2+1 \overline{k^4+3k^3+2k^2+3k+1} \\ \underline{k^4 \phantom{+3k^3} + k^2} \phantom{+3k+1} \\ 3k^3+k^2+3k \phantom{+1} \\ \underline{3k^3 \phantom{+k^2} + 3k} \phantom{+1} \\ k^2 \phantom{+3k} + 1 \\ \underline{k^2 \phantom{+3k} + 1} \\ 0 \end{array}$$

$$\therefore \text{LHS} = \tan^{-1} \frac{2k+3}{-k^2-3k-1}$$

$$= \text{RHS}$$

Step 4: So proof holds by the inductive process